

Java Developer Roadmap (Basic to Advanced)

1. Core Java (Basic Level)

- Java Fundamentals: Data types, Variables, Operators, Loops, Arrays, Methods
- OOP Concepts: Class, Object, Inheritance, Polymorphism, Abstraction, Encapsulation
- String Handling: String, StringBuilder, StringBuffer
- Exception Handling: try-catch, throw, throws, Custom Exceptions
- Collections Framework: List, Set, Map, Iterator
- Java 8 Features: Lambda, Stream API, Functional Interface, Optional

2. Intermediate Java

- Multithreading: Thread, Runnable, Synchronization, Executor Framework
- File Handling: File class, BufferedReader/Writer, Serialization
- JDBC: Connection, Statement, PreparedStatement, ResultSet, Transactions
- Basic SQL: CRUD operations, JOIN, Index, Constraints

3. Advanced Java

- Servlets and JSP
- Spring Framework: Spring Core, Spring Boot, Spring MVC, Spring Data JPA
- REST API Development
- Hibernate / JPA
- Maven or Gradle
- Git and GitHub

4. Extra Skills

- Design Patterns
- SOLID Principles
- Microservices
- Basic Linux Commands
- Docker (Optional)
- Unit Testing: JUnit, Mockito

Job Ready Skills

- Build CRUD Applications
- Database Connectivity
- Create REST APIs
- Proper Exception Handling
- Version Control using Git
- Strong OOP Understanding